# Unit-V Tutorial

1. $H_0$ : The die is unbiased ; $H_1$ : The die is biased

Z-test : $z = \frac{x-\mu}{\sigma}$ , $n = 9000$ , $x = 3220$

$p$ = probability of getting 3 or 4 = $1/3$

$q = 1-p = 2/3$ ; $\mu = np = 9000 \times \frac{1}{3} = 3000$

$\sigma = \sqrt{npq} = \sqrt{3000 \times 2/3} = 20\sqrt{5}$

$$z = \frac{3220 - 3000}{20\sqrt{5}} = 4.92$$

At 1% LoS , $|Z_\alpha| = 2.58$ , $|z| > |Z_\alpha|$

$\therefore$ $H_0$ is rejected , the die is biased.

2. $n_1 = 32$ $\quad$ $n_2 = 36$ $\quad$ $H_0 : \bar{x}_1 = \bar{x}_2$

$\bar{x}_1 = 72$ $\quad$ $\bar{x}_2 = 70$ $\quad$ $H_1 : \bar{x}_1 \neq \bar{x}_2$

$s_1 = 8$ $\quad$ $s_2 = 6$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{72-70}{\sqrt{\frac{8^2}{32} + \frac{6^2}{36}}} = 1.1547$$

$|z_\alpha|$ at 1% LoS = 2.58

$\therefore |z| < |z_\alpha|$, $H_0$ is accepted

Hence, both boys & girls perform equally well.

③ A B

$\bar{x}_1 = 48$ $\quad \bar{x}_2 = 52$ $\quad H_0 : \bar{x}_1 = x_2$

$s_1 = 9$ $\quad s_2 = 14$ $\quad H_1 : \bar{x}_1 \neq x_2$

$n_1 = 450$ $\quad n_2 = 100$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{48-52}{\sqrt{\frac{9^2}{450} + \frac{14^2}{100}}}$$

$|z| = 2.7343$, At 5% LoS, $|z_\alpha| = 1.96$

$|z| > |z_\alpha|$, $H_0$ is rejected, there is significant difference.

④ $H_0 : \bar{x} = \mu$

$H_1 : \bar{x} \neq \mu$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{0.53 - 0.5}{0.03} \times 3 = 3$$

$\bar{x} = 0.53$ $\quad \nu = 10 - 1 = 9 = $ d.o.f

$\mu = 0.5$ i) At 1% LoS, $|t_\alpha| = 3.25$; $t < t_\alpha$, $H_0$ accepted

$s = 0.03$ ii) At 5% LoS $|t_\alpha| = 2.622$, $t > t_\alpha$, $H_0$ is rejected

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(5) $n_1 = 8$ $\quad n_2 = 7$ $\quad H_0 : \bar{x}_1 = \bar{x}_2$

$\bar{x}_1 = 17$ $\quad \bar{x}_2 = 16$ $\quad H_1 : \bar{x}_1 \neq \bar{x}_2$

$$s_1^2 = \frac{\Sigma(x_1 - \bar{x}_1)^2}{n} = \frac{1}{8}\left[2^2 + 0^2 + (-2)^2 + (4)^2 + (-1)^2 + 1^2 + 2^2 + 3^2\right]$$
$$= 4.5$$

$S_1 = \sqrt{S_1^2} = 2.1213$

$$s_2^2 = \frac{1}{7}\left[1^2 + 2^2 + 1^2 + 3^2 + 1^2 + 1^2\right] = 2.8571$$

$s_2 = 1.6903$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = 0.93$$

for $\nu = 13$ d.o.f., At 5% LoS :- $t_\alpha = 2.16$

$|t| < |t_\alpha|$

$\therefore H_0$ is accepted, they don't significantly differ.

(6) $n_1 = 19$ $\quad n_2 = 16$

$\Sigma(x_i - \bar{x})^2 = 150$, $\quad \Sigma(y_i - \bar{y})^2 = 99$

$H_0 : \sigma_1^2 = \sigma_2^2$ $\quad s_1^2 = \frac{\Sigma(x_i - \bar{x})^2}{n_1}$

$H_1 : \sigma_1^2 \neq \sigma_2^2$

$\Rightarrow s_1^2 n_1 = \Sigma(x_i - \bar{x})^2 = 160$

and $s_2^2 n_2 = \Sigma(y_i - \bar{y})^2 = 91$

$$F = \frac{\frac{n_1 s_1^2}{n_1 - 1}}{\frac{n_2 s_2^2}{n_2 - 1}} = \frac{\frac{160}{8}}{\frac{91}{7}} = 1.5385$$

At 5% LoS, $\nu = (9-1, 8-1) = (8,7)$

$\therefore F < F_\alpha$, $H_0$ is accepted

Hence they're drawn from the same normal population.

(7) $H_0 : \sigma_1^2 = \sigma_2^2$

$H_1 : \sigma_1^2 \neq \sigma_2^2$

$n_1 = 6$

$s_1^2 = \frac{\Sigma x_i^2}{n} - \bar{x}^2 = 1713 - 1708.4444$

$= 4.5556$

$n_2 = 7$

$s_2^2 = \frac{\Sigma y_i^2}{n} - \bar{y}^2 = 1711.5714 - 1704.5202 = 7.0612$

$$F = \frac{\frac{n_1 s_1^2}{n_1 - 1}}{\frac{n_2 s_2^2}{n_2 - 1}} = \frac{\frac{6(4.5556)}{5}}{\frac{7(7.0612)}{6}} = \frac{5.4667}{8.2381}$$

$= 1.507$

$\nu = (6,5)$ ; $F_\alpha = 4.95$

$F < F_\alpha$ ; $H_0$ is accepted

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8. $H_0$: Supports the result
$H_1$: Does not support the result

$O_1 = 220$, $O_2 = 170$, $O_3 = 90$, $O_4 = 20$

$O_1 + O_2 + O_3 + O_4 = 500 = \Sigma O_i$

$e_1 = 500 \times \frac{4}{10} = 200$ ; $\Sigma e_i = 500 = N$

$e_2 = 500 \times \frac{3}{10} = 150$

$e_3 = 500 \times 2/10 = 100$ ; $\chi^2 = \frac{\Sigma (O_i - e_i)^2}{e_i}$

$e_4 = 500 \times 1/10 = 50$

$$= \frac{(220-200)^2}{200} + \frac{(170-150)^2}{150} + \frac{(90-100)^2}{100} + \frac{(20-50)^2}{50}$$

$= 23.6667 \simeq 23.67$

d.o.f $= \nu = n-1 = 4-1 = 3$ ; $\chi^2_\alpha$ at 5% LoS $= 7.815$

Here $\chi^2 > \chi^2_\alpha$ ; $H_0$ is rejected

$\therefore$ The result is not supported.

9. $H_0$: The dice is fair
$H_1$: The dice aren't fair

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The probabilities of getting sum 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 & 12

| $X = x_i$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $P(x_i)$ | $1/36$ | $2/36$ | $3/36$ | $4/36$ | $5/36$ | $6/36$ | $5/36$ | $4/36$ | $3/36$ | $2/36$ | $1/36$ |

| Sum | $O_i$ | $E_i = 360 \cdot P(x_i)$ | $(O_i - E_i)^2$ | $\frac{(O_i - E_i)^2}{E_i}$ |
|---|---|---|---|---|
| 2 | 8 | 10 | 4 | 0.4 |
| 3 | 24 | 20 | 16 | 0.8 |
| 4 | 35 | 30 | 25 | 0.833 |
| 5 | 37 | 40 | 9 | 0.225 |
| 6 | 44 | 50 | 36 | 0.72 |
| 7 | 65 | 60 | 25 | 0.417 |
| 8 | 51 | 50 | 1 | 0.02 |
| 9 | 42 | 40 | 4 | 0.1 |
| 10 | 26 | 30 | 16 | 0.53 |
| 11 | 14 | 20 | 36 | 1.8 |
| 12 | 14 | 10 | 16 | 1.6 |
| $\Sigma$ | 360 | 360 | | 7.445 |

$\therefore \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 7.445$ ; $\nu = n - 1 = 11$

At 5% LoS, $\chi^2_\alpha = 18.3$ ; $\therefore \chi^2 < \chi^2_\alpha$,

$H_0$ is accepted & the dice are fair.

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(10) $H_0$: Experiment supports the theory
$H_1$: Experiment does not support the theory

$O_1 = 315,\ O_2 = 108,\ O_3 = 101,\ O_4 = 32$

$\Sigma O_i = 556$

$e_1 = 556 \times \frac{9}{16} = 312.75$

$e_2 = 556 \times \frac{3}{16} = 104.25$

$e_3 = 556 \times \frac{3}{16} = 104.25$

$e_4 = 556 \times \frac{1}{16} = 34.75$

$$\chi^2 = \frac{\Sigma(O_i - E_i)^2}{E_i} = \frac{(315-312.75)^2}{312.75} + \frac{(108-104.25)^2}{104.5} + \frac{(101-104.25)^2}{104.25} + \frac{(32-34.75)^2}{34.75} = 0.47$$

$\nu = n - 1 = 4 - 1 = 3$ d.o.f.

At 1% LoS, $\chi^2_\alpha = 11.3$
At 5% LoS, $\chi^2_\alpha = 7.81$

$\chi^2 < \chi^2_\alpha$; $H_0$ is accepted
Hence the exp. supports the theory at both LoS.